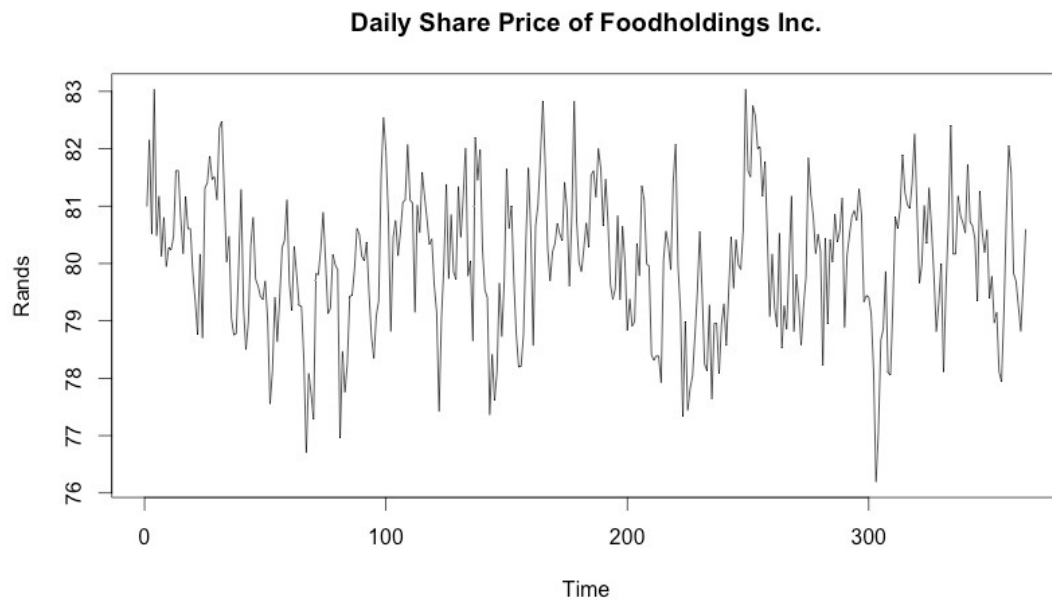


Week 12 Tutorial – Time Series

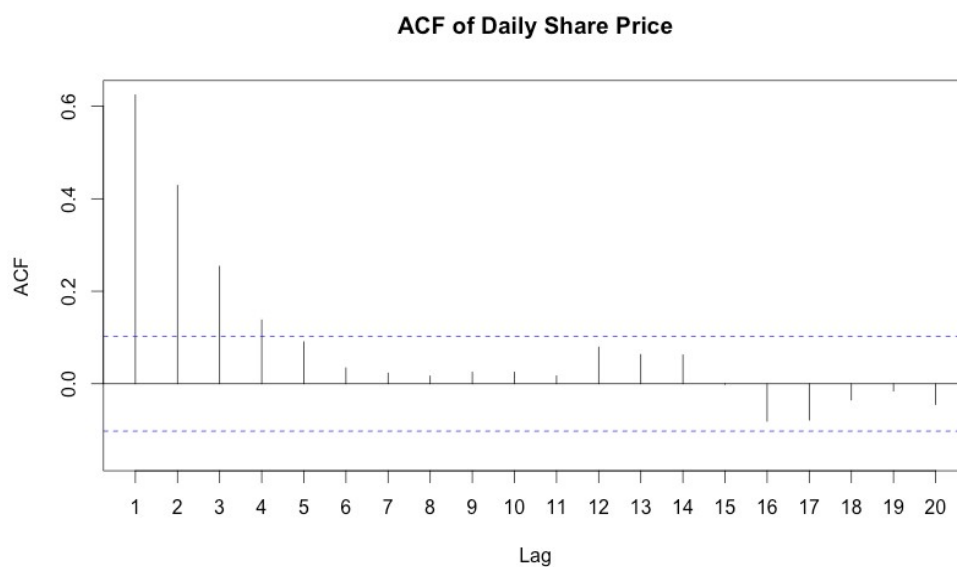
QUESTION 1

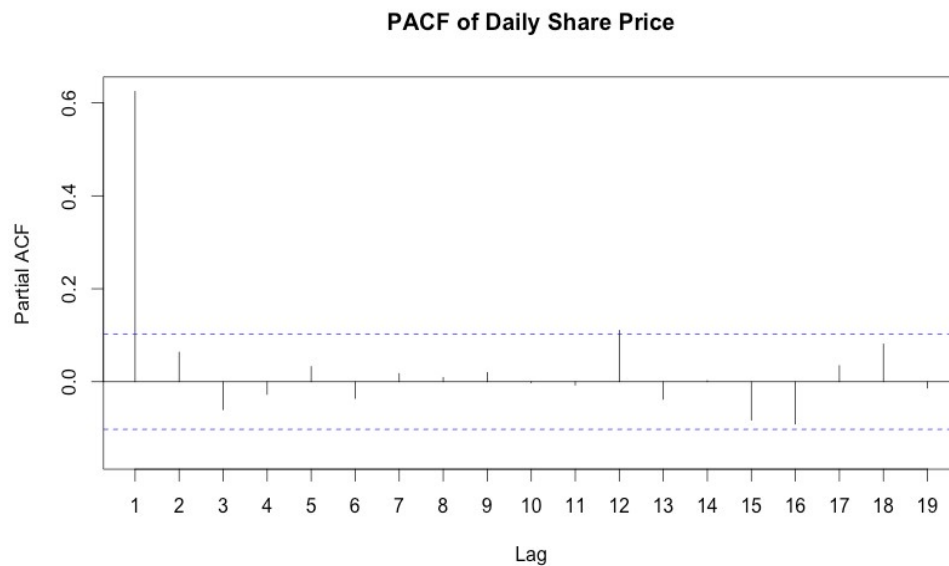
Consider the daily closing share price of Foodholdings Inc., a food recovery company. The data for 365 days have been plotted as a time series below.



a) Does this time series appear to be stationary? Justify your answer

The correlogram and partial correlogram for the daily closing share prices are plotted below.





- b) Is there autocorrelation present in the time series? Justify your answer by interpreting appropriate output.
- c) What data would you fit a model to?
- d) Based on your answer in (d) and the behaviour of the sample ACF and PACF plots, what type of model would you suggest fitting to the data? Justify your answer.
- e) Can you suggest the order of the model(s) in (e) that should be fit to the data?

A model was fit to the data, and the following output was obtained:

ARIMA(1,0,0)

Coefficients:

	<i>ar1</i>	<i>xmean</i>
	0.6244	80.0476
s.e.	0.0407	0.1326

sigma² estimated as 0.9136

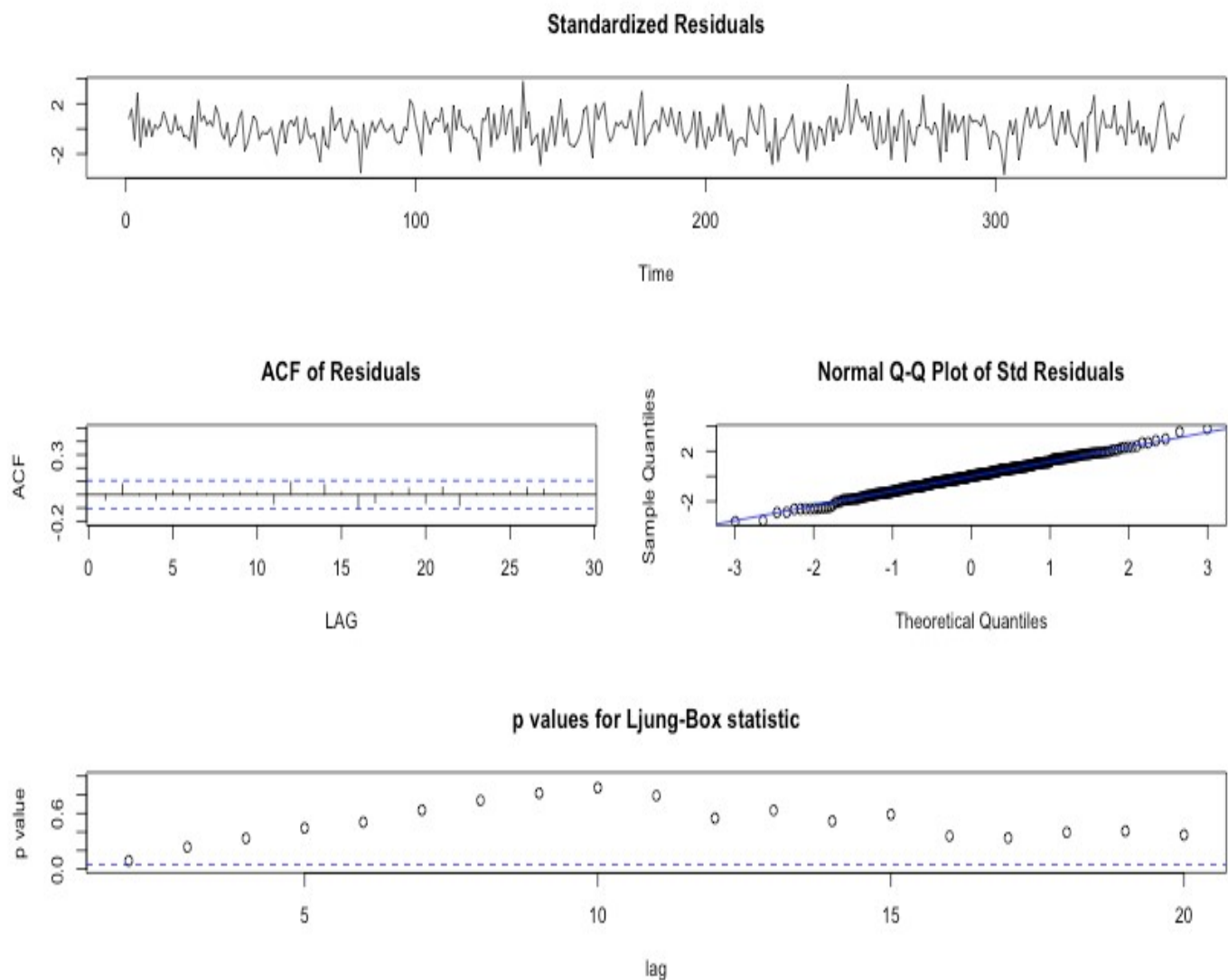
AIC=1009.35 AICc=1009.42 BIC=1021.05

- f) State the fitted model mathematically.
- g) What is the value of the constant term of the model?
- h) What is the estimated variance of the error/random noise series?

- i) Compute the variance and autocorrelation (at lag 1) of the fitted model. Show the formula that you use in your calculations.

Residual diagnostics were carried out on the fitted ARIMA(1,0,0) model, with the appropriate output displayed below.

- j) Why are the residual diagnostics carried out?
- k) What can you conclude about the model from the residual diagnostic output? Justify your answer by referring to each element of the output.



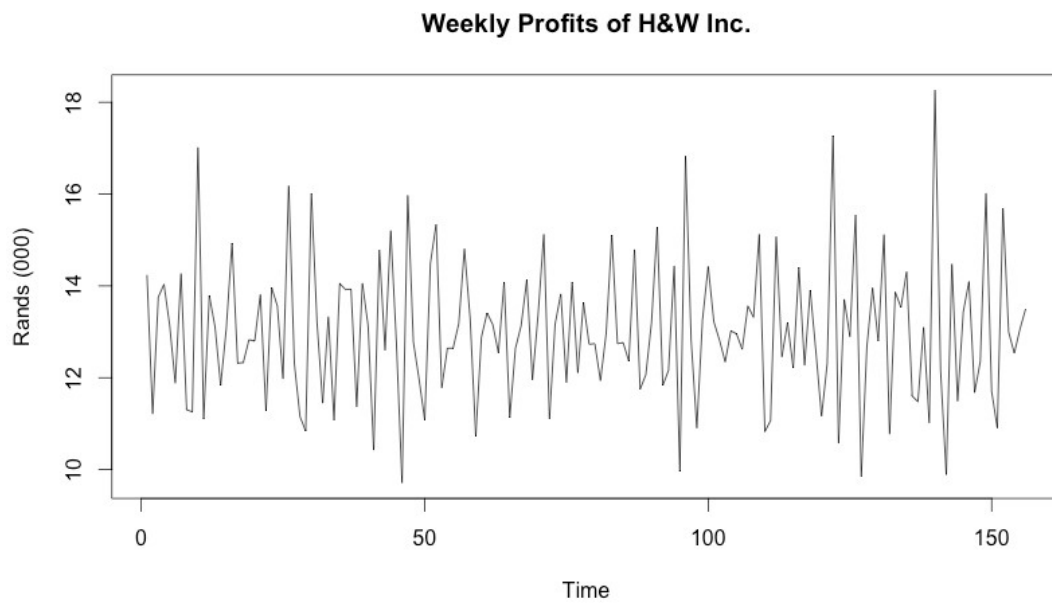
QUESTION 2

Suppose that an ARIMA(p, d, q) model is to be fitted to some time series data $\{Y_t\}_{t=1, \dots, T}$

- a) Describe what important features of the data can be revealed by a time series plot .
- b) Explain how the orders p , d and q can be identified.
- c) Having fitted an ARIMA(p, d, q) model to the data, what residual diagnostics should be looked at, and why?

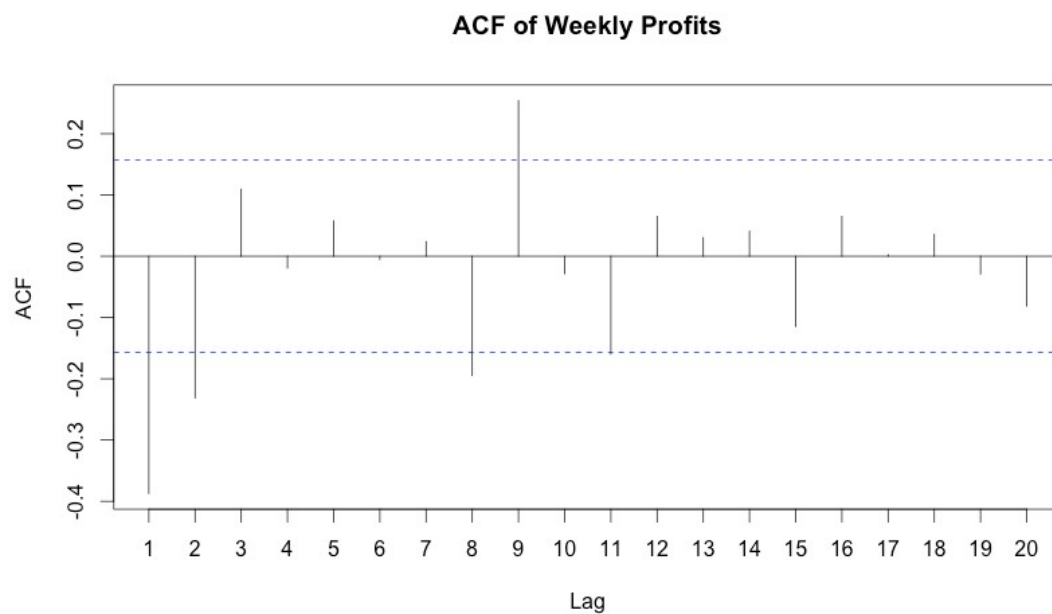
QUESTION 3

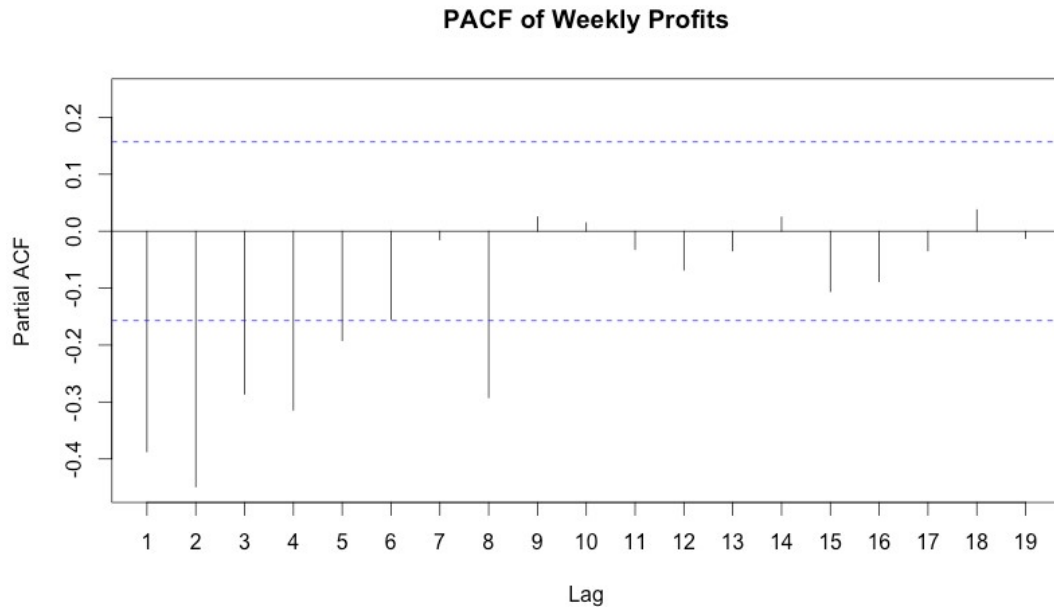
The weekly profits of Holmes&Watson Inc., a privately-owned detective agency that operates out of 221B Baker Street, London are plotted below.



a) Does the time series appear stationary? Justify your answer.

The sample ACF and PACF plots of the original data are provided below





b) Based on the above output, can you suggest an appropriate type of model to fit to the data?

c) Based on your answer in (b), what order of model should be fit?

Four mixed autoregressive-moving-averages models were fit to the data, along with their AIC values, displayed in the table below.

<u>Model</u>	<u>ARIMA(1,0,1)</u>	<u>ARIMA(2,0,1)</u>	<u>ARIMA(1,0,2)</u>	<u>ARIMA(2,0,2)</u>
AIC	490.62	483.61	487.29	486.83

d) Explain what information criteria measure, and how they can be used in order to guide the order selection of a model.

e) Which would be the best model with which to proceed in this case? Justify your answer.

f) What would be the next step of the general ARIMA-modelling approach? Describe how you would perform the step and what conclusion you would look to make at the end of the step.

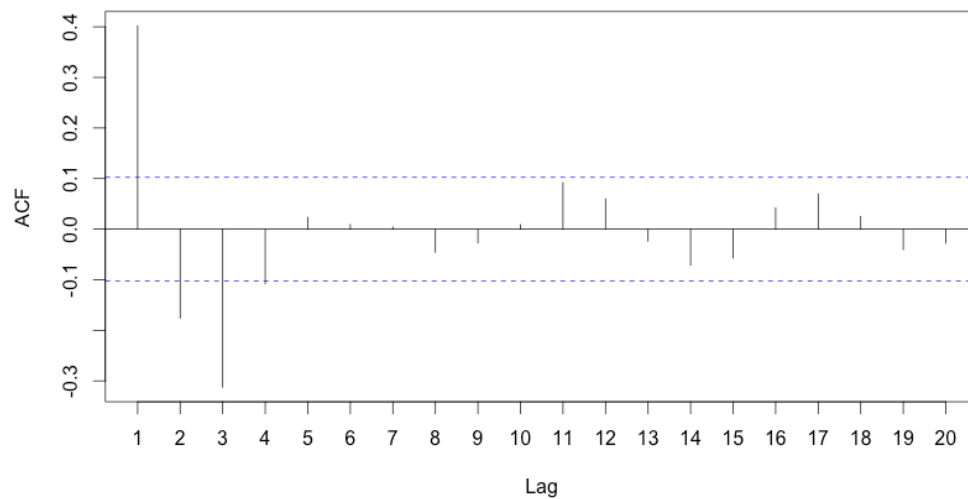
QUESTION 4

For each of the following examples:

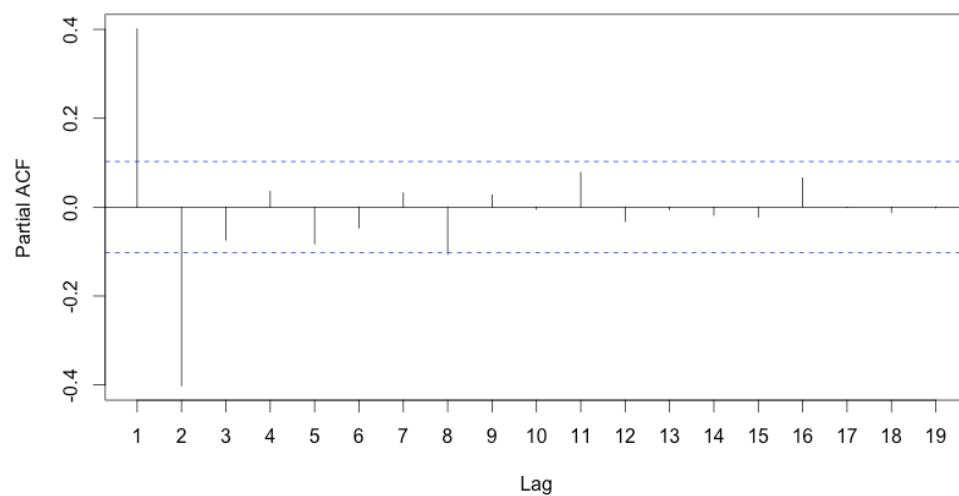
- i) Describe the behaviour of the sample ACF plot
- ii) Describe the behaviour of the sample PACF plot
- iii) Suggest an appropriate model/s to fit to the data if the ACF and PACF plots are from a stationary original time series
- iv) Suggest an appropriate model/s to fit to the data if the ACF and PACF plots are from the transformed data of a non-stationary time series that has been differenced once to make it stationary

A

ACF

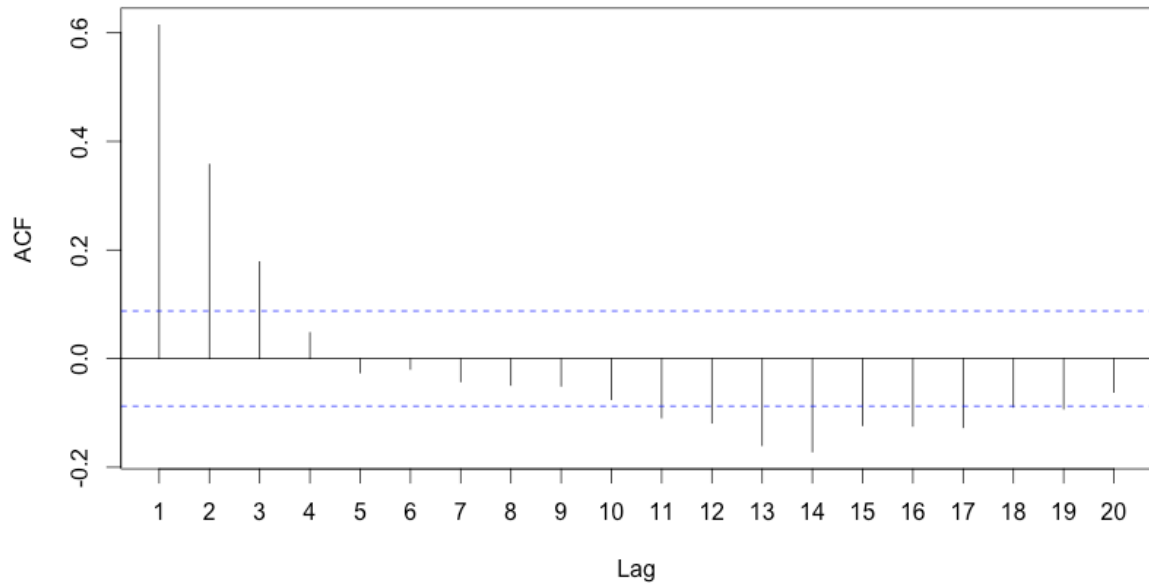


PACF

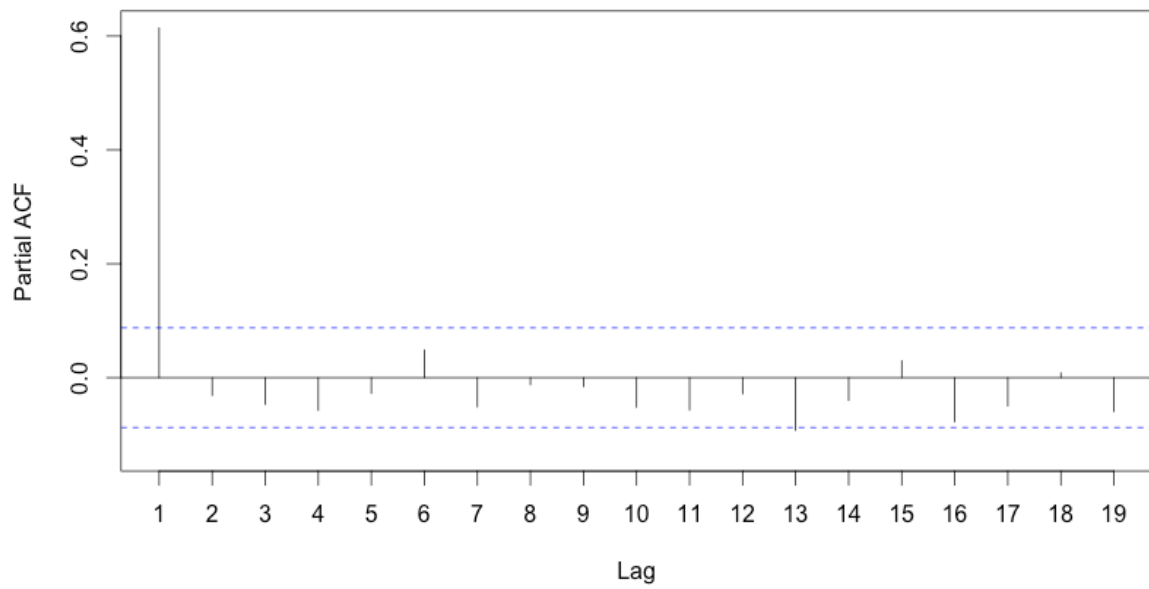


B

ACF

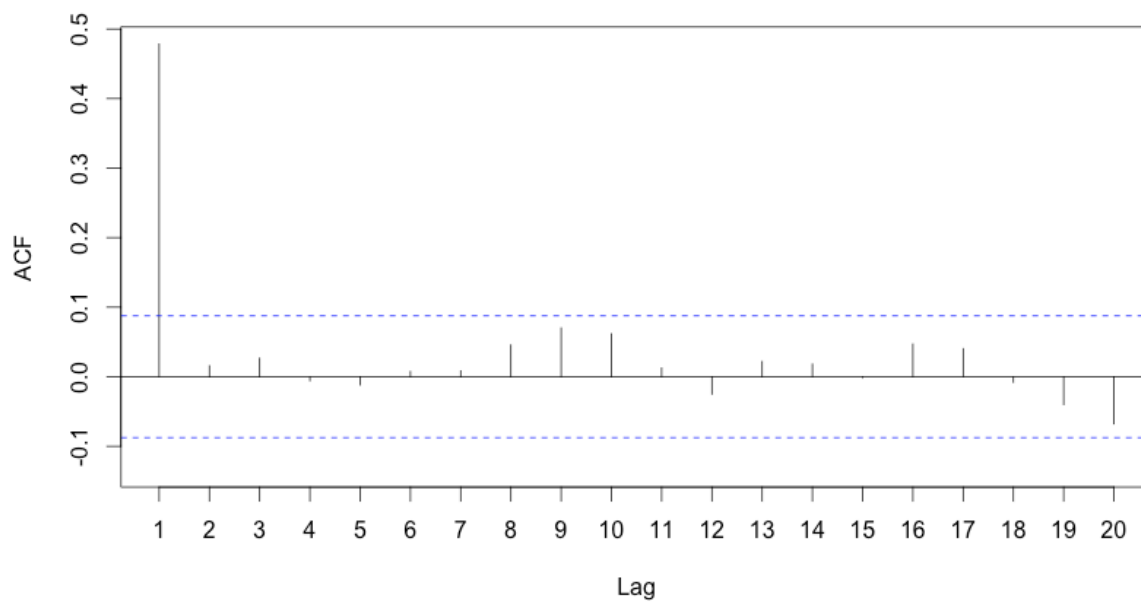


PACF

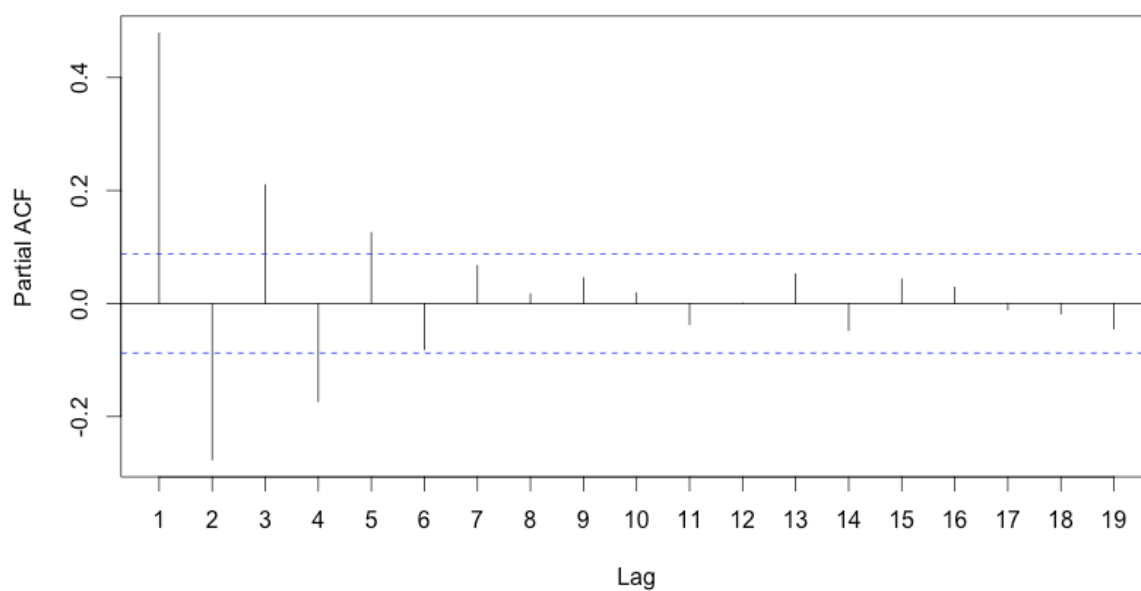


C

ACF

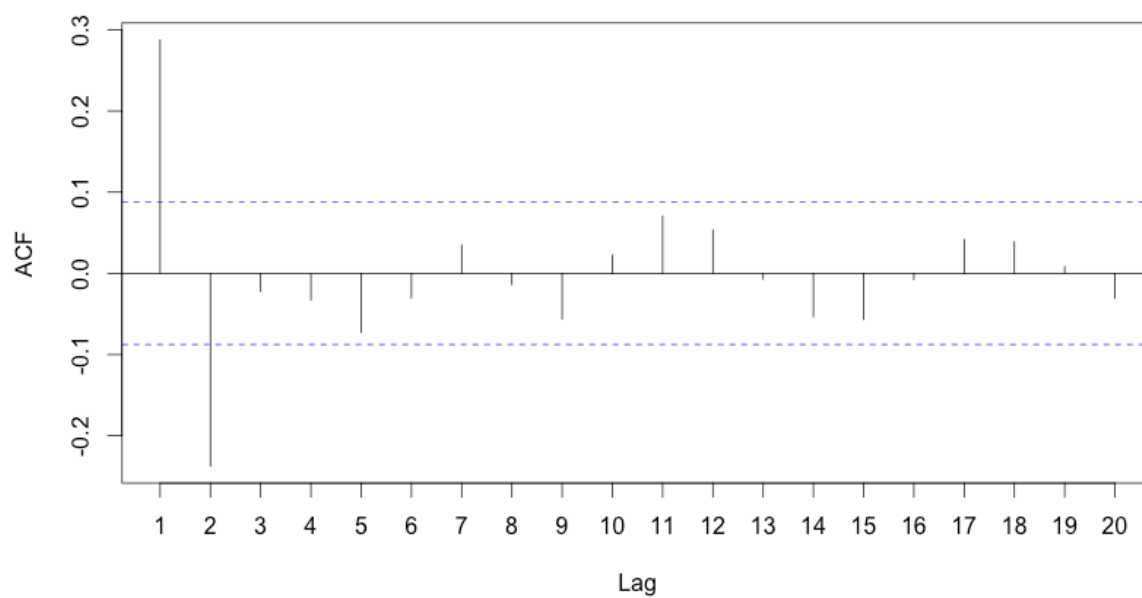


PACF

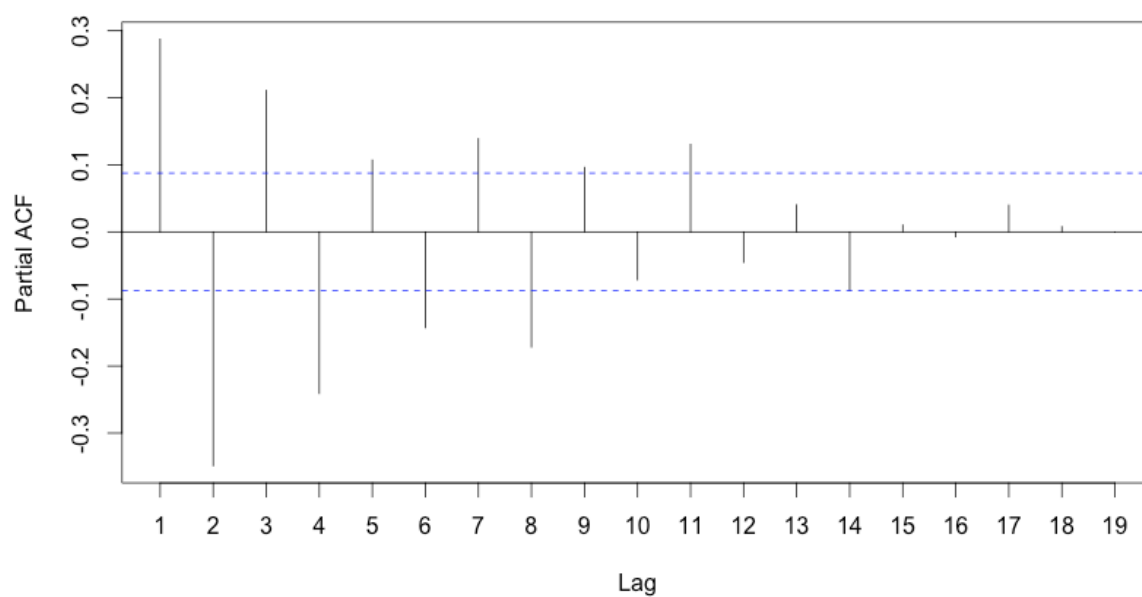


D

ACF

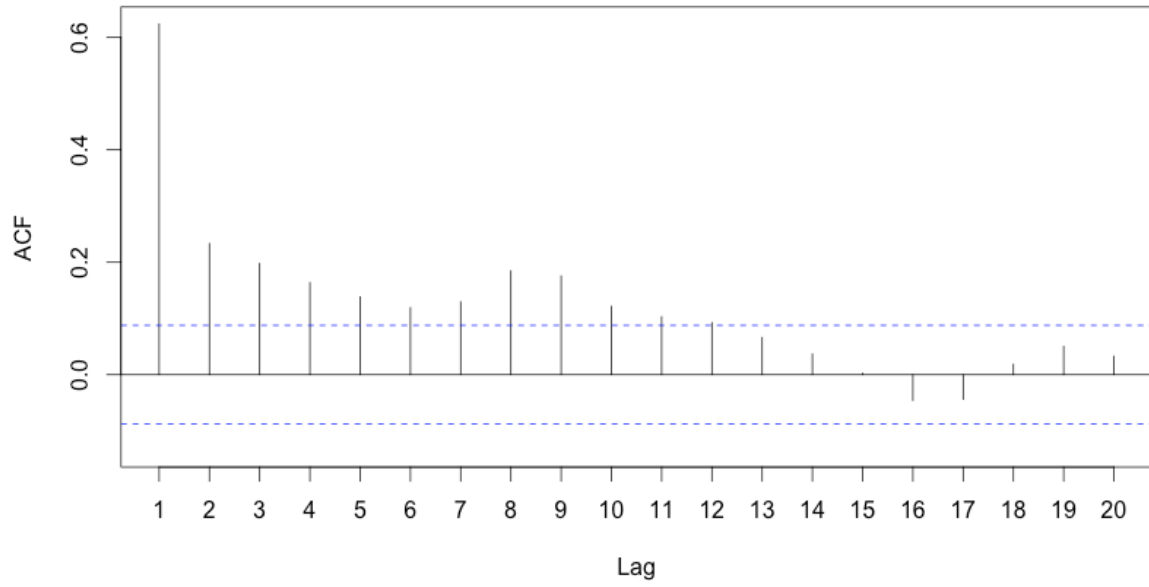


PACF

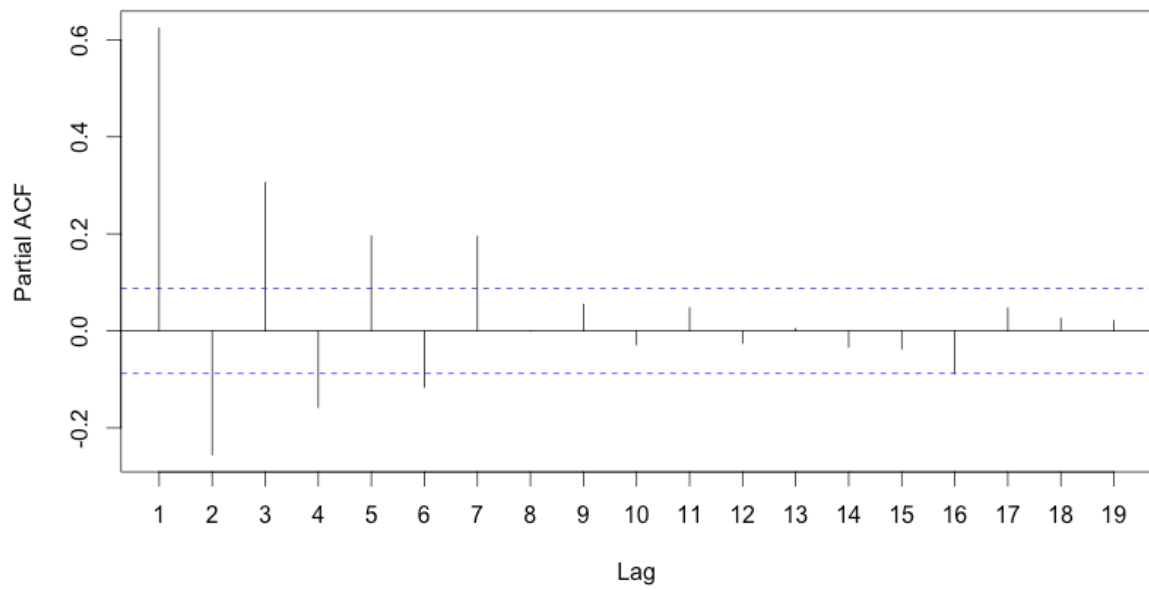


E

ACF

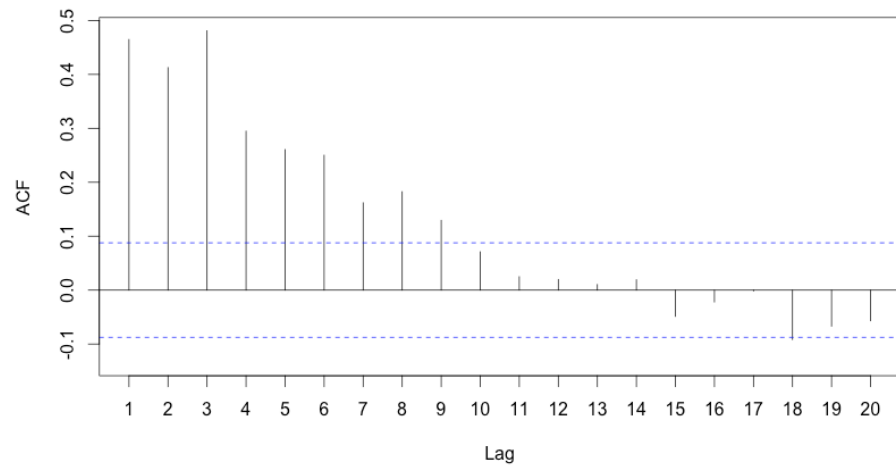


PACF

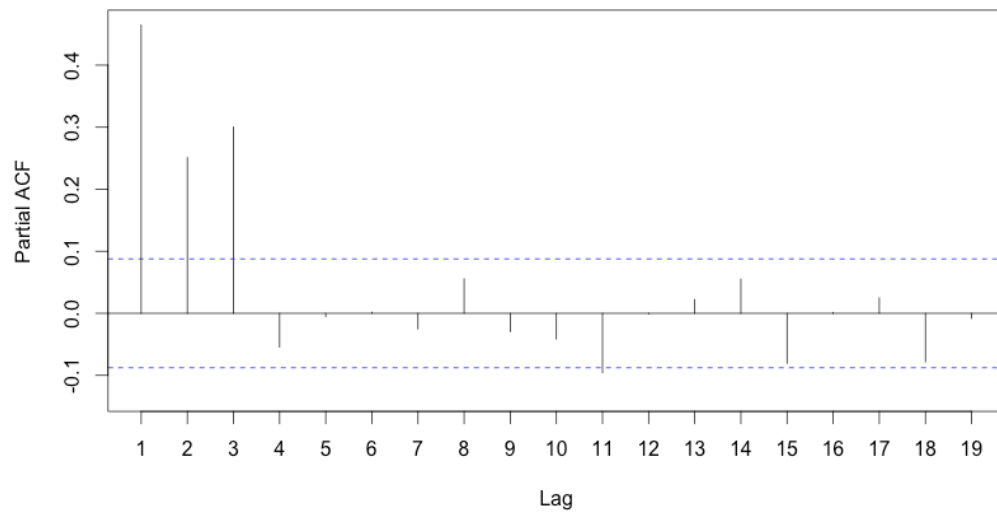


F

ACF



PACF



QUESTION 5

For each of the following examples:

- i) Identify the type of process (e.g. AR(2); MA(1))
- ii) Determine the mean of the process
- iii) Determine the value of the autocorrelation at lag $p = 1$
- iv) Determine the variance of the process

A

$$X_t = 0.62X_{t-1} + \epsilon_t$$

where

$$E[\epsilon_t] = 0 \text{ and } V[\epsilon_t] = 2.21$$

B

$$X_t = 22.45 + \epsilon_t - 0.49\epsilon_{t-1}$$

where

$$E[\epsilon_t] = 0 \text{ and } V[\epsilon_t] = 1.01$$

QUESTION 6

Consider the time series plot of a subset of the dataset 'ausbeer', the Total quarterly beer production in Australia in megalitres.



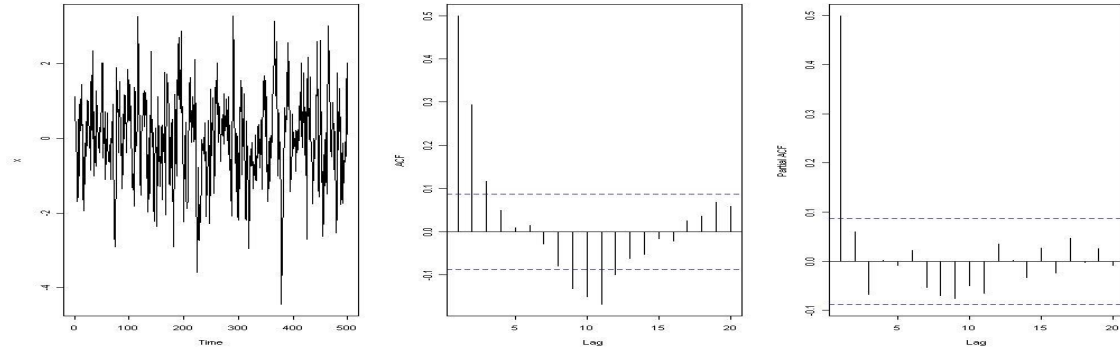
- What time series components are visible in this time series? Be sure to describe them fully.
- Why is the time series non-stationary? Justify your answer.
- We split the data into two sets – a 'training' set that contains around 80% of the data and a 'test' set that contains the remaining 20%. Why do we split the data up in this way?
- Based on the RStudio output below, describe how you would difference the training data to obtain a stationary time series?

```
> nsdiffs(train)
[1] 1
> ndiffs(train)
[1] 1
```

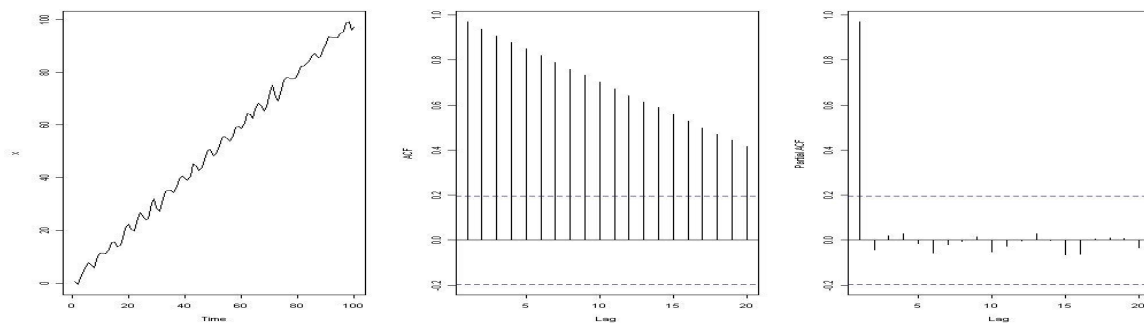
QUESTION 7

The following plots belong to several time series data. Each row gives the time series plot, sample autocorrelation function and sample partial autocorrelation function of one series.

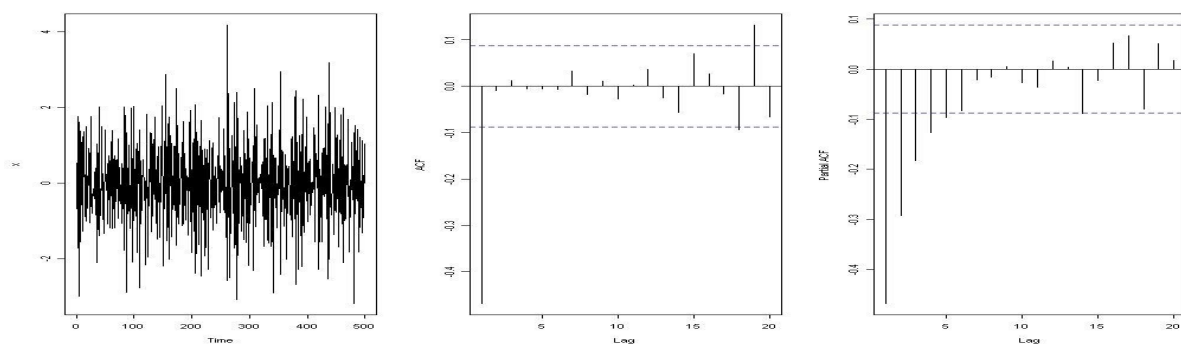
A



B



C



- Examine each time series and explain whether the series appears stationary or not.
- If the time series does appear to be stationary, what is the order of ARIMA model for that series? Explain how you obtain your answer. If the time series is not stationary, what potential transformations could be applied to the data in order to transform it into a stationary series of residuals?

QUESTION 8

Consider the first-order moving average process $\{X_t\}$ where

$$X_t = 8.372 + Z_t + 0.8Z_{t-1}.$$

and $\{Z_t\}$ is a white noise process with $E(Z_t) = 0$ and $\text{Var}(Z_t) = 5.312$

Obtain the mean and variance and autocorrelation at lag 1 of $\{X_t\}$, showing your workings.